K-TRACE AND K-DETERMINANT OF AN L-LINEAR MAP FACTOR THROUGH THE L-TRACE AND L-NORM

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The case E_1(a) uses the block decomposition of a : L -> L over a K-basis, whose K-determinant is by definition Norm_{L/K}(a) (context span).

Editorial note. The label lemma-trace-and-norm-linear names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 2596-2610: the definitions of field trace and norm as K-trace and K-determinant of the multiplication map, with K-linearity of the trace and multiplicativity of the norm
- Linear algebra: trace is additive and det multiplicative in the matrix; GL_n is generated by elementary matrices of the two displayed types

Definition 0.1. Let L/K be a finite extension of fields. For $\alpha \in L$ we define the *trace* $\mathrm{Trace}_{L/K}(\alpha) = \mathrm{Trace}_K(\alpha : L \to L)$ and the *norm* $\mathrm{Norm}_{L/K}(\alpha) = \det_K(\alpha : L \to L)$.

It is clear from the definition that $\mathrm{Trace}_{L/K}$ is K-linear and satisfies $\mathrm{Trace}_{L/K}(\alpha) = [L : K]\alpha$ for $\alpha \in K$. Similarly $\mathrm{Norm}_{L/K}$ is multiplicative and $\mathrm{Norm}_{L/K}(\alpha) = \alpha^{[L:K]}$ for $\alpha \in K$.

Lemma 0.2. *Let L/K be a finite extension of fields. Let V be a finite dimensional vector space over L. Let $\varphi : V \to V$ be an L-linear map. Then*

$$\mathit{Trace}_K(\varphi : V \to V) = \mathit{Trace}_{L/K}(\mathit{Trace}_L(\varphi : V \to V))$$

and

$$\det_K(\varphi : V \to V) = \mathit{Norm}_{L/K}(\det_L(\varphi : V \to V))$$

Proof. Choose an isomorphism $V = L^{\oplus n}$ so that φ corresponds to an $n \times n$ matrix. In the case of traces, both sides of the formula are additive in φ. Hence we can assume that φ corresponds to the matrix with exactly one nonzero entry in the (i, j) spot. In this case a direct computation shows both sides are equal.

In the case of norms both sides are zero if φ has a nonzero kernel. Hence we may assume φ corresponds to an element of $\mathrm{GL}_n(L)$. Both sides of the formula are

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multiplicative in φ. Since every element of $\mathrm{GL}_n(L)$ is a product of elementary matrices we may assume that φ either looks like

$$E_{12}(\lambda) = \begin{pmatrix} 1 & \lambda & \ldots \\ 0 & 1 & \ldots \\ \ldots & \ldots & \ldots \end{pmatrix} \quad \text{or} \quad E_1(a) = \begin{pmatrix} a & 0 & \ldots \\ 0 & 1 & \ldots \\ \ldots & \ldots & \ldots \end{pmatrix}$$

(because we may also permute the basis elements if we like). In both cases the formula is easy to verify by direct computation. $\square$